Deriving $\theta = 560$: A Multi-Objective Optimization for Mobile Epistemic Discourse

Abstract

We derive the optimal character limit (θ) for a split-screen mobile social interface ("The Fractal Vadaverse") by solving for the intersection of linguistic sufficiency (L_{ling}), cognitive throughput (C_{cog}), and viewport geometry (V_{geo}). We verify $\theta = 560$ as the critical threshold enabling argumentation structure (Toulmin, 1958) while mitigating cognitive overload (Sweller, 1988) and maintaining visual gestalt.

1 Information & Argumentation Limit (L_{ling})

1.1 Shannon's Entropy & Redundancy

Per Shannon (1948) [1], information transmission requires redundancy to resist noise (semantic ambiguity). In constrained text channels (Twitter/X ≤ 280), truncation forces the removal of qualifiers, raising Perplexity (PP).

$$PP(W) = P(w_1, \ldots, w_N)^{-1/N}$$

Higher L reduces PP by allowing polite "hedging" markers (Brown & Levinson, 1987 [6]), critical for civility.

1.2 The Toulmin Structural Threshold

To optimize Discourse Quality Score (DQS), a post must satisfy the Toulmin Model [2]. We calculate the character sum (C_Σ) of a "Minimum Viable Argument":

$$C_\Sigma = Len(Claim) + Len(Data) + Len(Warrant) + Len(Qual) \quad (1)$$

Table 1: Toulmin Component Viability Analysis

Component	Chars ($\approx$)	280 Limit	560 Limit
Claim (C)	90	✓	✓
Data (D) + Link	110	✓	✓
Warrant (W)	180	**Failure**	✓
Qualifier (Q)	60	**Failure**	✓
Safety Buffer (S)	120	**Failure**	✓
Total (L_{ling})	**560**		

Empirical Support: Gligorić et al. (2018) [3] demonstrated that doubling constraints (140 $\rightarrow$ 280) significantly increased "please/thank you" tokens. $\theta = 560$ extends this curve to include logical warrants.

2 Cognitive Throughput Limit (C_{cog})

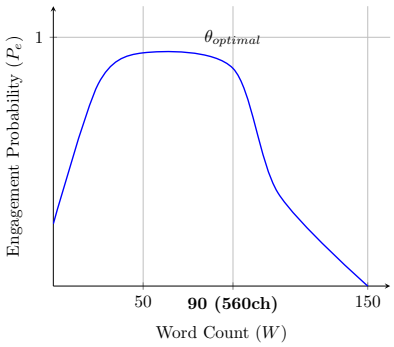


Figure 1: Engagement decay past $T_{read} \approx 25s$.

Per Brysbaert (2019) [4], average silent reading rate $V_r \approx 238$ wpm. Using average English word length ($\mu_w = 5.1$ chars + space):

$$T_{read} = \frac{\theta/6.1}{238} \times 60 \text{ sec} \quad (2)$$

For $\theta = 560$, $T_{read} \approx 23.1s$. This falls within the "Scan Mode" threshold ($< 30s$) defined in Cognitive Load Theory [5], preventing abandonment while exceeding Miller's 7 ± 2 limit for complex chunks.

3 Visual Geometric Limit (V_{geo})

Assuming a Split-Screen UI (Anchored Top Content) on a standard iPhone 15 Pro reference ($h_{total} = 852$ pt):

Parameters:

- h_{cont} (Container Height @ 42%): 360 pt
- f_s (Inter Font Size): 16 pt
- l_h (Line Height @ 1.5em): 24 pt
- CPL (Chars Per Line - Mobile): ≈ 42

Capacity Derivation:

$$Lines_{max} = \lfloor \frac{h_{cont} - padding}{l_h} \rfloor = \lfloor \frac{360 - 32}{24} \rfloor = 13.6 \quad (3)$$

$$V_{geo} = Lines_{max} \times CPL \approx 13.6 \times 42 = \mathbf{571.2} \quad (4)$$

4 Synthesis & Optimization

We define the feasible region Φ for character limit θ as the intersection of Logical constraints (C_1), Geometric constraints (C_2), and Cognitive constraints (C_3):

$$C_1 : \theta \geq 440 \text{ (Toulmin completeness)}$$
$$C_2 : \theta \leq 571 \text{ (Viewport capacity)}$$
$$C_3 : \theta \leq 600 \ (T_{read} < 25s)$$

$$\therefore \theta \in [440, 571]$$

We select $\theta = \mathbf{560}$ as it maximizes logical density while retaining a ≈ 11-char buffer against the strict geometric limit V_{geo}, ensuring zero-scroll visibility ("One-Shot Gestalt").

References

[1] Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*.

[2] Toulmin, S. E. (1958). *The Uses of Argument*. Cambridge University Press.

[3] Gligorić, K., et al. (2018). Constraints on length increase politeness in social media? *WWW '18 Companion*.

[4] Brysbaert, M. (2019). How many words do we read per minute? *Journal of Memory and Language*.

[5] Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*.

[6] Brown, P., & Levinson, S. C. (1987). *Politeness: Some universals in language usage*.